

Wenbo Lyu

B.Sc. Graduate

Curriculum Vitae

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Education and qualifications

B.Sc. In Geographic Information Science Shaanxi Normal University 2025

Research Experience

2026.6- Research Assistant, The Hong Kong Polytechnic University
2025.9-2026.6 Research Assistant, Eastern Institute of Technology, Ningbo
2024.8-2025.8 Research Intern, The Hong Kong University of Science and Technology (Guangzhou)

Research interests

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.

Refereed research papers

Note: Publications under "Lyu, W." and "Lv, W." both refer to Wenbo Lyu.

Grants

Where the data already lives in a data frame, the same options are given as cell options instead, and the cell prints a table for the filter to pick up. See the README for that form.

2019–2024 **Australian Research Council Discovery Grant**

- *Forecasting large collections of related time series*
- \$540,000

2015–2018 **Australian Research Council Linkage Grant**

- *Forecasting electricity demand at the household level*
- \$320,000